
AWS CodeBuild – Overview

What is AWS CodeBuild?

AWS CodeBuild is a **fully managed build service** in AWS that compiles source code, runs tests, and produces software packages that are ready to deploy.

It eliminates the need to manage your own build servers and scales automatically to handle multiple builds at once.

Core Features

Feature	Description
Managed Build Service	No need to provision or scale build servers.
Pay-as-you-go	Pay only for the compute time you use.
Automatic Scaling	Can process multiple builds in parallel.
Custom Build Environments	Use default environments or create your own Docker images.
CI/CD Integration	Integrates with CodePipeline, GitHub, Bitbucket, CodeCommit, etc.
Secure	Supports IAM for permissions and VPC for network isolation.

How CodeBuild Works

1. **Source Code** – Pulled from a repository (e.g., CodeCommit, GitHub, Bitbucket).
2. **Buildspec File** – A YAML file (`buildspec.yml`) that defines the build steps.
3. **Build Environment** – A Docker container where your build runs.
4. **Artifacts** – Output like binaries or zip files stored in S3.

5. **Logs** – Sent to Amazon CloudWatch Logs for monitoring.

Sample buildspec.yml

version: 0.2

phases:

install:

commands:

- echo Installing dependencies...
- npm install

build:

commands:

- echo Building the project...
- npm run build

artifacts:

files:

- '**/*'

You can also define environment variables, reports, and cache settings in this file.

Integrations

- **Source Control:** GitHub, GitHub Enterprise, Bitbucket, AWS CodeCommit
 - **CI/CD:** AWS CodePipeline, Jenkins, GitHub Actions
 - **Testing & Reports:** Integrates with CodeGuru, CodeCoverage, and custom test suites
 - **Notifications:** Through Amazon CloudWatch, SNS, or EventBridge
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Typical Use Cases

- Continuous Integration in DevOps pipelines
- Automated testing for every code commit or pull request

- Building Docker images (and pushing to Amazon ECR)
 - Creating deployment packages for Lambda, ECS, or EC2
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Security & Permissions

- Use **IAM roles** to control what CodeBuild can access.
 - Run builds in a **VPC** for access to internal resources.
 - Store secrets in **AWS Secrets Manager** or **Parameter Store**.
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Pricing

- Charged **per build minute**, based on compute type.
 - Free tier: **100 build minutes/month** on `build.general1.small`.
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Build Environment Options

Option	Use Case
Standard Docker images	Prebuilt images for most common languages
Custom Docker images	Hosted in Amazon ECR or DockerHub
Privileged mode	Required for Docker-in-Docker (e.g., building containers)

Benefits

- Fully managed and scalable
- Deep integration with other AWS DevOps services

- Supports custom build workflows
 - Efficient and cost-effective CI solution
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